Homework 3 – Regression

1 Question 1

Below, I simulate a rather quite annoying set of regression data.

```
set.seed(0510)
n <- 250
dat.q1 <- tibble(x1 = rnorm(n, mean = 50, sd = 10),
                 x2 = factor(sample(c("A", "B"), n, replace = TRUE)),
                 x3 = factor(sample(c("Control", "Treat"), n, replace = TRUE)),
                 x4 = factor(sample(c("Low", "High"), n, replace = TRUE), levels=c("Low", "High")),
                 x5 = rnorm(n, mean = 0, sd = 1),
                 x6 = factor(sample(c("Group1", "Group2"), n, replace = TRUE)),
                 x7 = rnorm(n, mean = 10, sd = 2),
                 x8 = rnorm(n, mean = 10, sd = 2)) |>
        mutate(y = 5 +                          # Intercept
                 # Quantiative x1
                 2 * x1 +
                 # Categorical x2
                 (x2 == "B") * 10 +
                 # Categorical x Categorical (x3*x4)
                 (x3 == "Treat") * 5 + (x4 == "High") * 2 +
                 (x3 == "Treat" & x4 == "High") * 15 +
                 # Quantitative x Categorical  (x5*x6)
                 3 * x5 + (x6 == "Group2") * 4 +
                 (x5 * (x6 == "Group2")) * 8 +
                 # Quantitative x Quantitative (x7*x8)
                 1.5 * x7 + 0.5 * x8 +
                 0.8 * (x7 * x8) +
                 # Error
                 rnorm(n, mean = 0, sd = 15))
```

1.1 Part a

Fit the regression model.

$$
\begin{aligned}
Y_i = &\beta_0 + \beta_1 x_{1i} + \beta_2 I(x_{2i} = \text{B}) + \beta_3 I(x_{3i} = \text{Treat}) + \beta_4 I(x_{4i} = \text{High}) + \\
&\beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_7 x_{7i} + \beta_8 x_{8i} + \\
&\beta_9 (I(x_{3i} = \text{Treat}) \times I(x_{4i} = \text{High})) + && \text{(Categorical} \times \text{Categorical)} \\
&\beta_{10} (x_{5i} \times I(x_{6i} = \text{Group2})) + && \text{(Continuous} \times \text{Categorical)} \\
&\beta_{11} (x_{7i} \times x_{8i}) + \epsilon_i && \text{(Continuous} \times \text{Continuous)}
\end{aligned}
$$

Solution: First, we fit the regression model using `lm()`.

```
mod.q1 <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.q1)
```

1.2 Part b

Conduct a residual and outlier analysis.

Solution:

Figure 1 shows that the distribution of residuals is roughly bell-shaped and the points fit on the line in the qq plot. Therefore, the residuals are normal. The histogram centers at 0, meaning the residuals have a mean of 0. From the scatterplot, we see the residuals spreading equally around 0, suggesting common variance.

In Figure 2, we can see that the dataset has several observations that have high leverage. All D_i are below 0.50, so we know that these observations are not strongly influential.

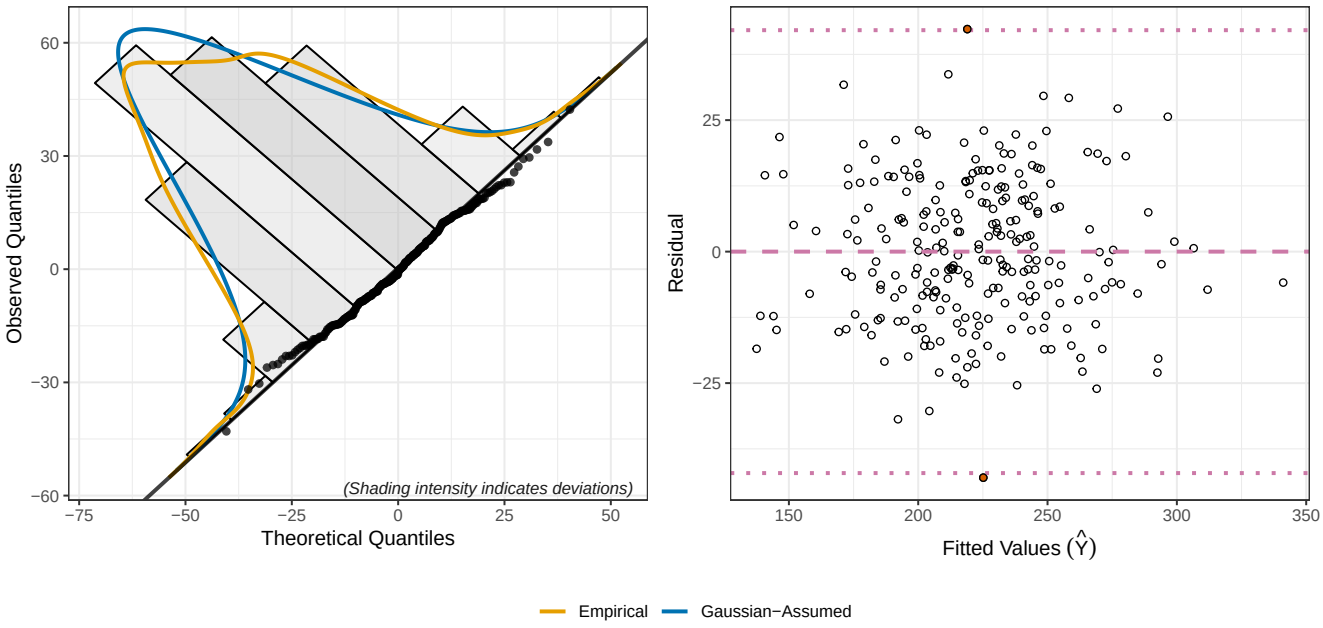
The scatterplot for the studentized residual and the DFFITs reveal some outliers. Based on the studentized residual plot, the residuals for 9 observations fall more than 2 SDs above or below 0. Regarding the DFFITs plot, our model has 12 predictors, so we compare the absolute value of DFFITs to $2 * \sqrt{12/250} = 0.44$ and $3 * \sqrt{12/250} = 0.66$. An observation between the 100th and the 125th observation is strongly influential, consistent with the results from studentized residuals calclation.

Overall, the data satisfies the conditions for use of regression regarding residual and outliers.

```
source("plotResiduals.R")
source("plotInfluence.R")
```

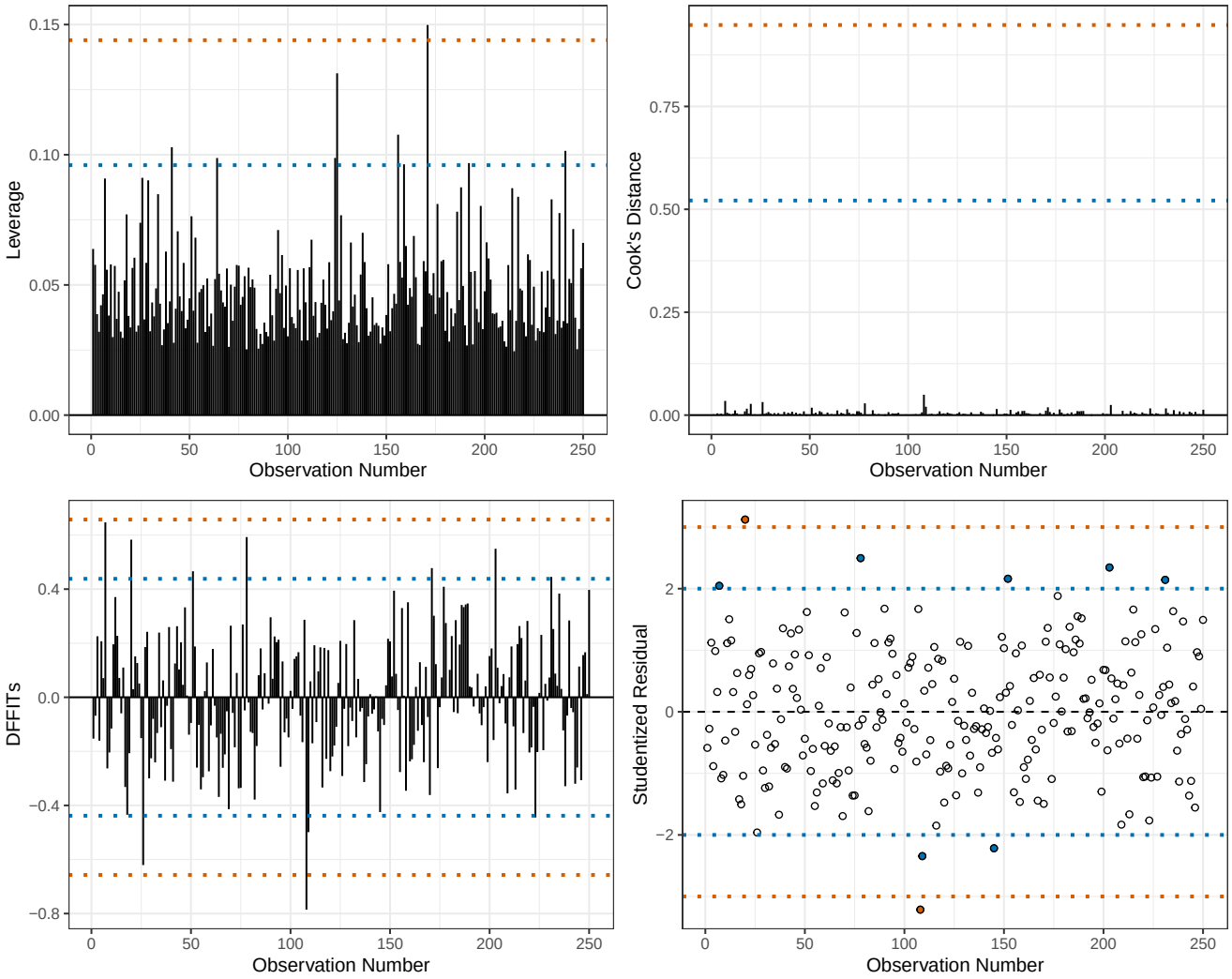
```
plotResiduals(mod.q1)
```

Figure 1: Residual Analysis




```
plotInfluence(mod.q1)
```

Figure 2: Outlider Analysis



1.3 Part c

Interpret the intercept, the coefficients on x_1 and x_2.

Solution:

- β_0 is the expected value of Y when every other predictor is at their reference value.
- β_1 is the expected increase in the value of Y with every 1 unit increase in x_1, given all other predictors are fixed.
- β_2 is the expected increase in the value of Y when the reference category for x_2 changes from A to B, given all other predictors are fixed.

1.4 Part d

Untangle and interpret the $x_3 * x_4$ interaction using algebra. Use Table 1 and Figure 3 for context.

Solution:

We hold all parts that do not involve x_3 and x_4 in the model constant, labeled as C. Our regression model becomes:

$$Y_i = \beta_3 I(x_{3i} = \text{Treat}) + \beta_4 I(x_{4i} = \text{High}) + \beta_9 (I(x_{3i} = \text{Treat}) \times I(x_{4i} = \text{High})) + C$$

Our expected Y, conditioned on x_3 and x_4, are as follow:

$$E(Y_i|x_{3i} = \text{Control}, x_{4i} = \text{Low}) = C$$

$$E(Y_i|x_{3i} = \text{Control}, x_{4i} = \text{High}) = \beta_4 + C$$

$$E(Y_i|x_{3i} = \text{Treat}, x_{4i} = \text{Low}) = \beta_3 + C$$

$$E(Y_i|x_{3i} = \text{Treat}, x_{4i} = \text{High}) = \beta_3 + \beta_4 + \beta_9 + C$$

- C is the expected value of Y in the Control group when condition is Low.
- β_3 is the expected difference Y between the Control and Treatment group when the condition is Low. In other words, in the Low condition, β_3 is the expected increase in Y when the reference category changes from Control to Treatment.
- β_4 is the expected difference Y between the Low and High condition in the Control group.
- β_9 is the interaction effect of Group (Treatment vs. Control) and Condition (High vs. Low). When group changes from Control to Treatment and condition changes from Low to High, the expeced value of Y gets both the effect of each categorical variable and the interaction effect.

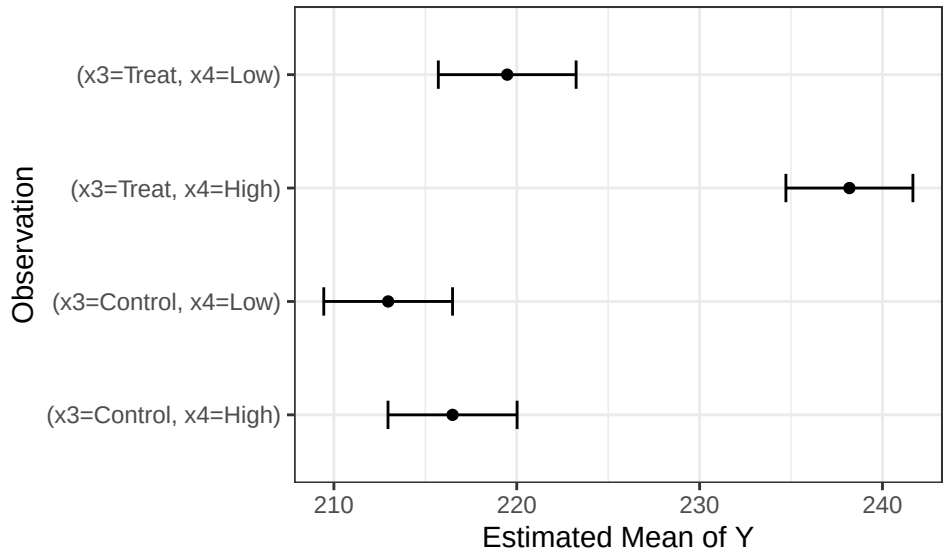
```r
library(emmeans)
emm.q1d <- emmeans(mod.q1, ~x3*x4) |> # Don't average across levels: use |x2*x6
            data.frame() |>
            rename("Mean"  = "emmean",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL")
```

Table 1: Estimated marginal means for the model in Question 1.

x3	x4	Mean	SE	df	Lower	Upper
Control	Low	212.9676	1.7871	238	209.4470	216.4882
Treat	Low	219.4792	1.9123	238	215.7120	223.2464
Control	High	216.4894	1.7916	238	212.9601	220.0188
Treat	High	238.1930	1.7631	238	234.7198	241.6662

```r
emm.q1d <- emm.q1d |>
  mutate(Observation = paste("(x3=", x3, ", ", "x4=", x4, ")", sep=""))
ggplot(emm.q1d) +
  geom_point(aes(x=Mean, y=Observation))+
  geom_errorbar(aes(y=Observation,
                    xmin=Lower, xmax=Upper),
                width = 0.25)+
  # facet_wrap(~x2*x6)+
  theme_bw()+
  xlab("Estimated Mean of Y")
```

Figure 3: Estimated marginal means for the model in Question 1.



1.5 Part e

Untangle and interpret the $x_5 * x_6$ interaction using algebra. Use Table 2 and Figure 4 for context.

Solution:

We hold all parts that do not involve x_5 and x_6 in the model constant, labeled as C. Our regression model becomes:

$$Y_i = \beta_5 x_{5i} + \beta_6 I(x_{6i} = \text{Group2}) + \beta_{10}(x_{5i} \times I(x_{6i} = \text{Group2})) + C$$

Our expected Y, conditioned on x_5 and x_6, are as follow:

- Group 1:
$$Y_i = \beta_5 x_{5i} + C$$
  - C is the expected value of Y when $x_5 = 0$.
  - β_5 is the increase in the expected value of Y when x_5 increases by 1 unit.
- Group 2:
$$Y_i = (\beta_5 + \beta_{10}) x_{5i} + (\beta_6 + C)$$
  - $\beta_6 + C$ is the expected value of Y when $x_5 = 0$.
  - $\beta_5 + \beta_{10}$ is the increase in the expected value of Y when x_5 increases by 1 unit. β_{10} is the difference between the slope of Group 1 and Group 2.

```
emm.q1e <- emtrends(mod.q1, var="x5", ~x5|x6) |>
            data.frame() |>
            rename("Slope"  = "x5.trend",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL") |>
            dplyr::select(x6, Slope, SE, df, Lower, Upper)
```

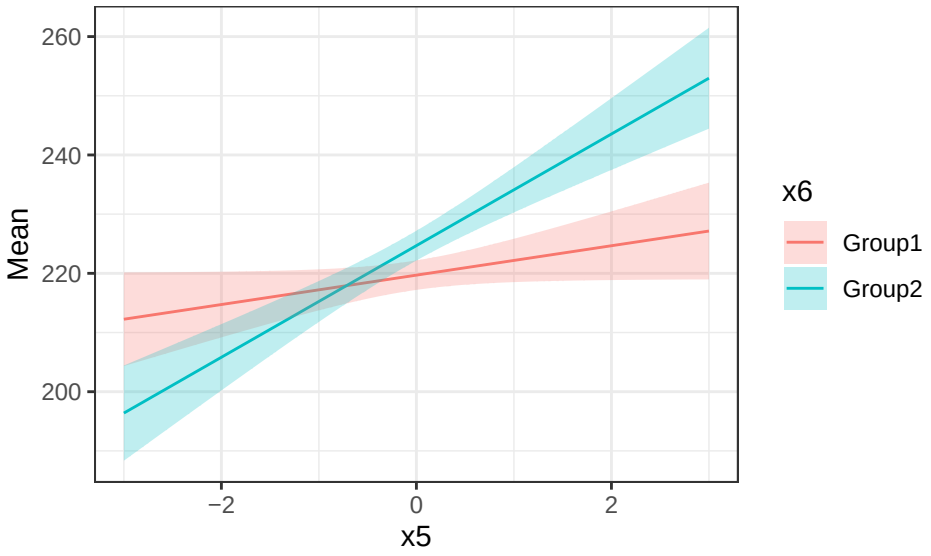
Table 2: Estimated marginal trends for the model in Question 1.

x6	Slope	SE	df	Lower	Upper
Group1	2.4835	1.2942	238	-0.0660	5.0330
Group2	9.4309	1.3367	238	6.7977	12.0642

```
emm.q1e <- emmeans(mod.q1, ~x5*x6,
                   at = list(x5 = seq(-3, 3, length.out=500))) |> # Don't average across levels: use |x2*x3*x4
  data.frame() |>
            rename("Mean"  = "emmean",
                   "Lower" = "lower.CL",
                   "Upper" = "upper.CL")

ggplot(emm.q1e) +
  geom_line(aes(x=x5, y=Mean, color=x6))+
  geom_ribbon(aes(x=x5, ymin=Lower, ymax=Upper, fill=x6), alpha=0.25)+
  # facet_wrap(~x2*x3*x4)+
  theme_bw()
```

Figure 4: Estimated marginal trends for the model in Question 1.



1.6 Part f

Untangle and interpret the $x_7 * x_8$ interaction using algebra. Use Table 3 and Table 4 for context.

Solution:

We hold all parts that do not involve x_7 and x_8 in the model constant, labeled as C. Our regression model becomes:

$$Y_i = \beta_7 x_{7i} + \beta_8 x_{8i} + \beta_{11}(x_{7i} \times x_{8i}) + C$$

We can combine like terms to get:

$$Y_i = (\beta_7 + \beta_{11} x_{8i}) x_{7i} + \beta_8 x_{8i} + C$$
$$Y_i = (\beta_8 + \beta_{11} x_{7i}) x_{8i} + \beta_7 x_{7i} + C$$

- $\beta_7 + \beta_{11} x_{8i}$ is the change in Y for a unit change in x_7 after adjusting for the simultaneous linear change in x_8.
- $\beta_8 + \beta_{11} x_{7i}$ is the change in Y for a unit change in x_8 after adjusting for the simultaneous linear change in x_7.
- β_{11} is the change in the **slope of** x_7 when x_8 increases by 1 unit, and is also the change in the slope of x_8 when x_7 increases by 1 unit. From the tables, we can see that increases in x_8 is associated with increases in the slope of x_7, and vice versa.
- When $x_8 = 0$, β_7 is the change in the expected value of Y when x_7 increases by 1 unit.
- When $x_7 = 0$, β_8 is the change in the expected value of Y when x_8 increases by 1 unit.

```
emm.q1f.x7 <- emtrends(mod.q1, var="x7", ~x7|x8,
                       at = list(x8 = mean(dat.q1$x8) + c(-1, 0 ,1)*sd(dat.q1$x8))) |>
              data.frame() |>
              rename("Slope"  = "x7.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 3: Estimated marginal trends for the model in Question 1.

x8	Slope	SE	df	Lower	Upper
8.1309	7.9230	0.6751	238	6.5932	9.2529
10.0582	9.7129	0.5015	238	8.7250	10.7008
11.9854	11.5027	0.7396	238	10.0458	12.9597

```r
emm.q1f.x8 <- emtrends(mod.q1, var="x8", ~x8|x7,
                       at = list(x7 = mean(dat.q1$x7) + c(-1, 0 ,1)*sd(dat.q1$x7))) |>
              data.frame() |>
              rename("Slope"  = "x8.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 4: Estimated marginal trends for the model in Question 1.

x7	Slope	SE	df	Lower	Upper
8.3824	6.8569	0.6069	238	5.6613	8.0525
10.2208	8.5643	0.4744	238	7.6297	9.4989
12.0592	10.2716	0.7325	238	8.8286	11.7146

A Johnson Neyman analysis using the `interactions` package for `R` (Long 2024), shows that the slope of x_7 is statistically discernible for all observed values of x_8 (Figure 6), and vice versa (Figure 5).

```r
library(interactions)
johnson_neyman(mod.q1, pred = x7, modx = x8, control.fdr = T)
```

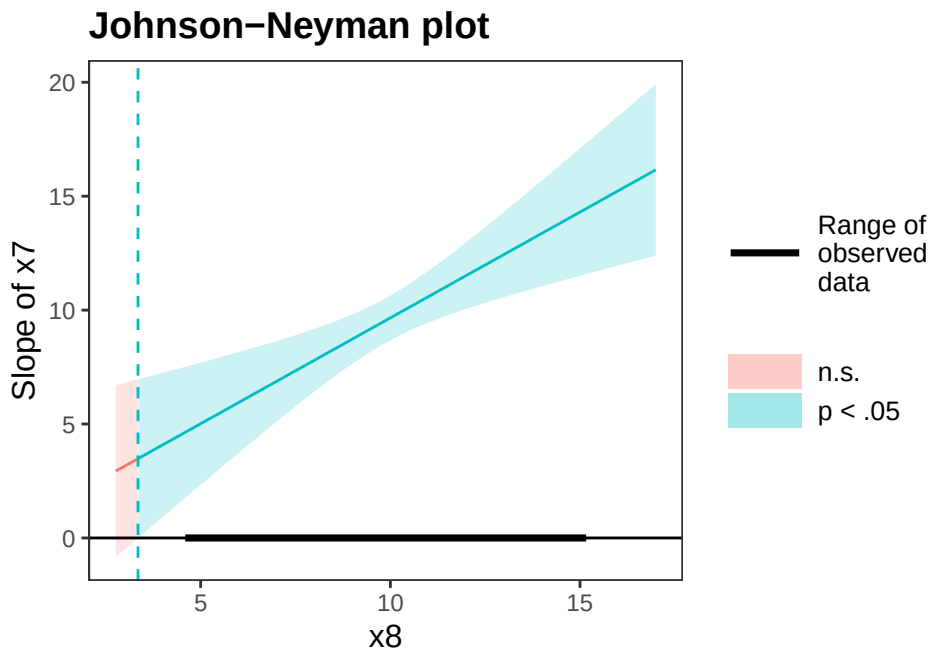
```
JOHNSON-NEYMAN INTERVAL

When x8 is OUTSIDE the interval [-13.08, 3.35], the slope of x7 is p < .05.

Note: The range of observed values of x8 is [4.69, 15.07]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 5: Johnson Neyman interval for the model in Question 1.




```r
johnson_neyman(mod.q1, pred = x8, modx = x7, control.fdr = T)
```

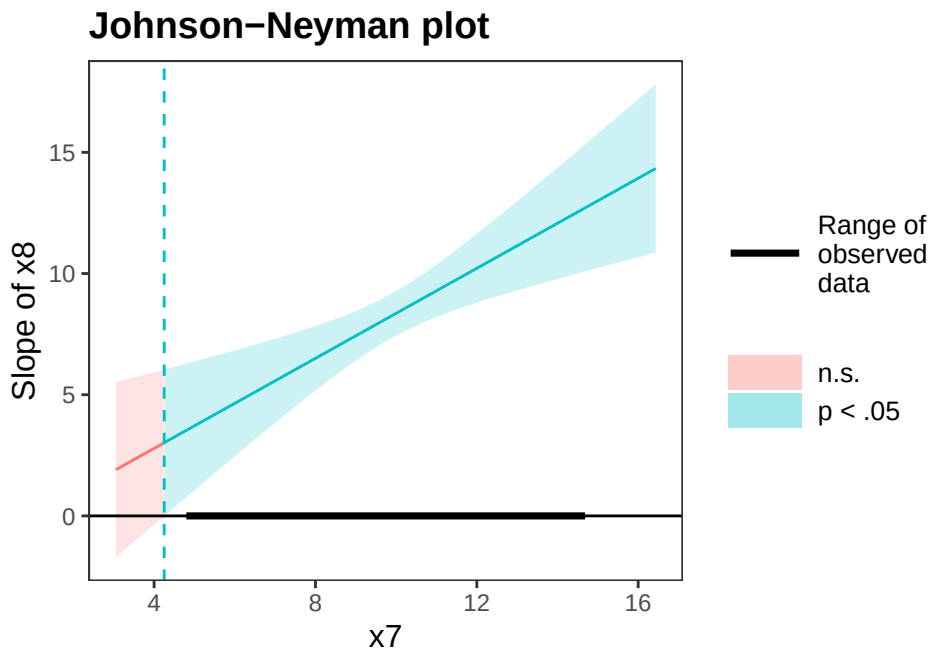
```
JOHNSON-NEYMAN INTERVAL

When x7 is OUTSIDE the interval [-9.96, 4.25], the slope of x8 is p < .05.

Note: The range of observed values of x7 is [4.89, 14.60]

Interval calculated using false discovery rate adjusted t = 1.97
```

Figure 6: Johnson Neyman interval for the model in Question 1.



1.7 Part g

Standardize the continuous data and refit the regression model. Redo part f. What do you notice about the coefficient estimates and why does this make the untangling process easier?

Solution: First, we fit the regression model using `lm()`.

```
# Standardize Data
dat.z <- dat.q1 |>
  mutate(across(c(x1, x5, x7, x8), ~ as.vector(scale(.))))
# Refit Model
mod.q1z <- lm(y ~ x1 + x2 + x3*x4 + x5*x6 + x7*x8, data=dat.z)
```

```
emm.q1g.z7 <- emtrends(mod.q1z, var="x7", ~x7|x8,
                    at = list(x8 = c(-1, 0 ,1))) |>
          data.frame() |>
          rename("Slope"  = "x7.trend",
                 "Lower" = "lower.CL",
                 "Upper" = "upper.CL") |>
          dplyr::select(x8, Slope, SE, df, Lower, Upper)
```

Table 5: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x8	Slope	SE	df	Lower	Upper
-1	14.57	1.24	238	12.12	17.01
0	17.86	0.92	238	16.04	19.67
1	21.15	1.36	238	18.47	23.83

```
emm.q1g.z8 <- emtrends(mod.q1z, var="x8", ~x8|x7,
                       at = list(x7 = c(-1, 0 ,1)))|>
              data.frame() |>
              rename("Slope"  = "x8.trend",
                     "Lower" = "lower.CL",
                     "Upper" = "upper.CL") |>
              dplyr::select(x7, Slope, SE, df, Lower, Upper)
```

Table 6: Estimated marginal trends for the model in Question 1 when quantitative variables are standardized.

x7	Slope	SE	df	Lower	Upper
-1	13.22	1.17	238	10.91	15.52
0	16.51	0.91	238	14.70	18.31
1	19.80	1.41	238	17.01	22.58

After transformation,

- β_7 is the effect of a 1-SD change in x_7 when x_8 is at its mean.
- β_8 is the effect of a 1-SD change in x_7 when x_7 is at its mean.
- β_{11} now becomes the change in the slope of 1 standardized predictor when the other predictor shifts by 1 standard deviation. For example, looking at the effect of x_7:

$$Y_i = (\beta_7 + \beta_{11} z_{8i}) z_{7i} + \beta_8 z_{8i} + C$$

- When $z_8 = -1$,

$$Y_i = (\beta_7 - \beta_{11}) z_{7i} - \beta_8 + C$$

  In other words, when x_8 is 1 SD below the means, for every increase in the SD of x_7, the expected value of Y increases by $\beta_7 - \beta_{11}$. When x_7 is at average, the expected value of Y is $C - \beta_8$. This transformation makes the untangling process easier. In Table 5, in the slope column, the first row is the estimated value for $\beta_7 - \beta_{11}$, the second row is β_7, and third row is the estimated value for $\beta_7 + \beta_{11}$. The difference between rows is the estimated value for β_{11}.

2 Question 2

Cui and Mao (2025) investigate how freezing and cryogenic temperatures affect the strength, flexibility, stiffness, and internal breaking behavior of 3D-printed CF/PEEK composites.

Carbon fiber reinforced PEEK (CF/PEEK) is a high-performance composite used in 3D printing for aerospace, deep-sea, polar research. The researchers aimed to assess whether and how extremely cold environments impact this composite to understand how 3D-printed CF/PEEK performs in freezing conditions. This work would be important to understand its limits and develop preventative methods to protect against structural failures.

The researchers created fifteen 3D printed dog-bone-shaped test specimens and placed them in a chamber cooled with liquid nitrogen to 25°C, −25°C, −75°C, −125°C, and −175°C. To test strength of the specimen, a machine pulled both ends of the specimen until it snapped.

In the table below, the researchers record the average tensile strength for three specimens from each temperature level.

Temperature	Mean Tensile Strength
25°C	81.93 MPa
−25°C	76.63 MPa
−75°C	63.23 MPa
−125°C	32.13 MPa
−175°C	12.83 MPa

2.1 Part a

Compute the estimates we derived using calculus in lecture 16:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_{1i} - \bar{x}_1)(y_i - \bar{y})}{\sum_{i=1}^{n} (x_{1i} - \bar{x}_1)^2}.$$

Solution:

$\beta_0 = 80.755$, $\beta_1 = 0.3654$

```
# create dataframe
df = tibble(
  x1 = c(25, -25, -75, -125, -175),
  y1 = c(81.93, 76.63, 63.23, 32.13, 12.83)
)
```

```r
# create function
func = function(x, y) {
  x_bar = mean(x)
  y_bar = mean(y)
  numerator = sum((x - x_bar) * (y - y_bar))
  denominator = sum((x - x_bar)^2)
  numerator / denominator
}
```

```r
# input dataframe into function
(b1 = func(df$x1, df$y1))
```

```
[1] 0.3654
```

```r
# calculate b0
(b0 = mean(df$y1) - b1 * mean(df$x1))
```

```
[1] 80.755
```

2.2 Part b

Compute the estimates we derived using linear algebra in lecture 16:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

Hint: Run `?matrix()` in `R` for a quick reminder about how to create a matrix.

Solution:

```r
# create matrices
x = matrix(data = c(1, 1, 1, 1, 1, 25, -25, -75, -125, -175),
           nrow = 5,
           ncol = 2)

y = matrix(data = c(81.93, 76.63, 63.23, 32.13, 12.83),
           nrow = 5,
           ncol = 1)
# compute beta hat
(bh = solve(t(x) %*% x) %*% t(x) %*% y)
```

```
        [,1]
[1,] 80.7550
[2,]  0.3654
```

2.3 Part c

Compute the maximum likelihood estimates using `optim()` in `R` as described in lecture 16:

$$\max_{\beta_0,\beta_1,\sigma} \mathcal{L}(\theta = \{\beta_0, \beta_1, \sigma\}).$$

Solution:

We know that $\epsilon_i \sim Gaussian(0, \sigma)$. We want to find the parameters (intercept, slope, and standard deviation of residual terms) that maximize the likelihood of observing our observed residual terms.

```r
# create function (optim can only adjust 1 object, so we have to put all 3 parameters into 1 vector)

neg_log = function(par, data, mle = F) {
  b0 = par[1]
  b1 = par[2]
  sigma = par[3]

  x = data$x1
  y = data$y1

  ll = sum(dnorm(y-(b0+b1*x), 0, sigma, log = T)) #point estimation
  ifelse(mle, -ll, ll)
}
```

```r
mle = optim(fn = neg_log,
            par = c(mean(df$y1), 0, sd(df$y1)), #if slope is 0, b0 = mean(y)
            data = df,
            mle=T) #passed in for the function, not optim
(mle$par[1])
```

```
[1] 80.75468
```

```r
(mle$par[2])
```

```
[1] 0.3653821
```

2.4 Part d

Fit the model using `lm()` and plot the resulting model. Compare to the results from parts (a) through (c).

Solution:

Using `lm`, we obtain the following results for the intercept and the coefficient:

```
mod.q2 = lm(y1 ~ x1, data = df)
(coef(mod.q2)["(Intercept)"])
```

```
(Intercept)
     80.755
```

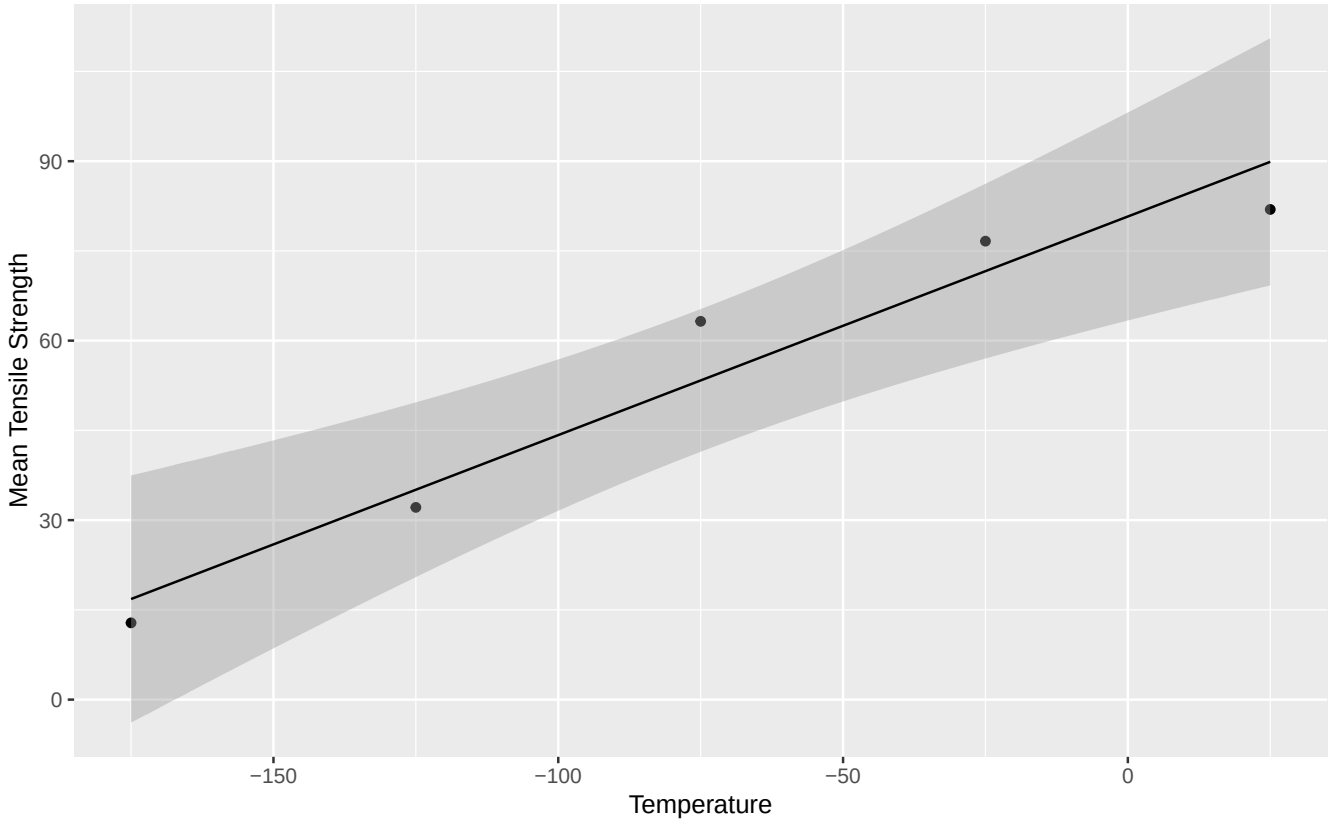
```
(coef(mod.q2)["x1"])
```

```
    x1
0.3654
```

The results are similar across from part (a) to (d).

```
ggplot(df, aes(x = x1, y = y1)) +
  geom_point(aes(x = x1,
                 y = y1)) +
  geom_smooth(aes(x = x1,
                  y = y1),
              method = "lm",
              color = "black",
              linewidth = 0.5) +
  labs(x = "Temperature",
       y = "Mean Tensile Strength")
```

Figure 7: Regression on Conditional Means of Y



3 Question 3

Bracht et al. (2016) evaluate MFAP4 as a biomarker for severity of fibrosis in hepatitic C patients.

```
dat.mfap4 <- read_csv("Bracht2016.csv")
```

3.1 Part a

Fit the model predicting `MFAP4` using `Fibrosis.Stage`. Write out the resulting estimate model and interpret the coefficients.

Solution:

$\hat{y}_i = 9.230 + 1.301 \times I(x_{1i} = \text{“Stage 1”}) + 3.154 \times I(x_{1i} = \text{“Stage 2”}) + 11.742 \times I(x_{1i} = \text{“Stage 3”}) + 15.388 \times I(x_{1i} = \text{“Stage 4”})$

- $\beta_0 = 9.230$: the expected value of Y when the reference category is “Stage 0”
- $\beta_1 = 1.301$: the change in Y when the reference category changes from “Stage 0” to “Stage 1”
- $\beta_2 = 3.154$: the change in Y when the reference category changes from “Stage 0” to “Stage 2”
- $\beta_3 = 11.742$: the change in Y when the reference category changes from “Stage 0” to “Stage 3”
- $\beta_4 = 15.388$: the change in Y when the reference category changes from “Stage 0” to “Stage 4”

```r
# transform dataset
dat = dat.mfap4 |>
  select(Fibrosis.Stage, MFAP4)

# fit model
mod.q3.1 = lm(MFAP4 ~ Fibrosis.Stage, data = dat)
```

3.2 Part b

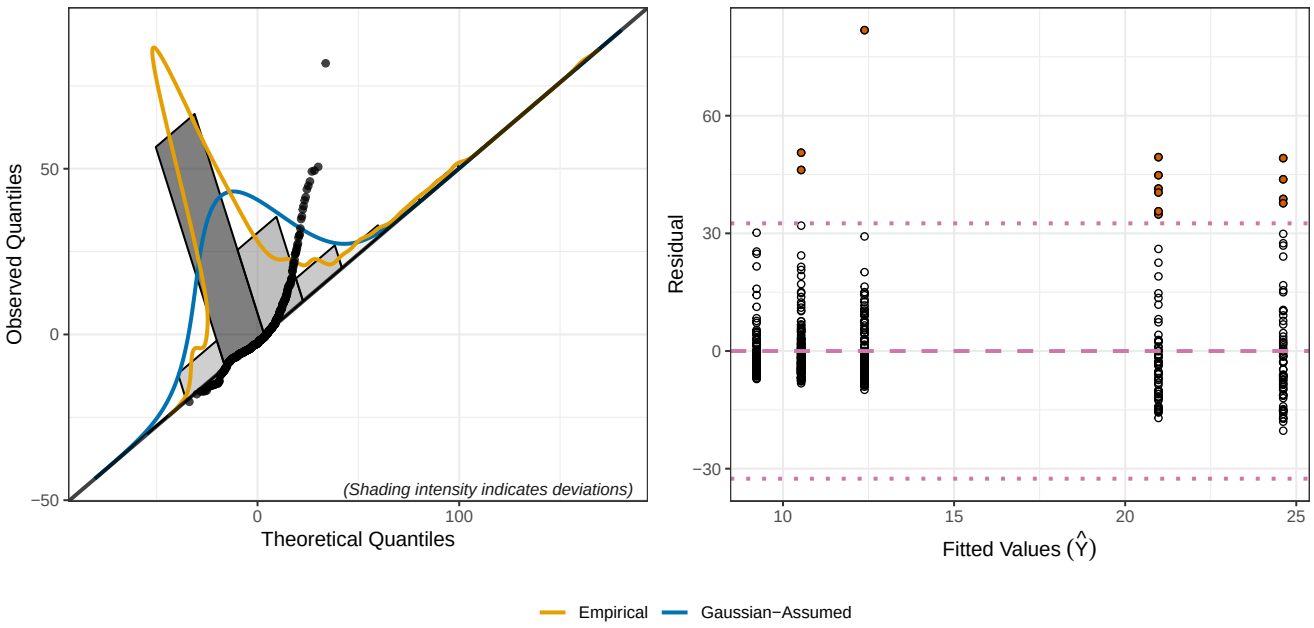
Conduct a residual analysis on the model in part a to demonstrate how to identify when you need a log transformation.

Solution:

From Figure 8, we can see that the residual terms are not normal, and a good amount of them are way oft the qq line. The variance follows a megaphone shape. For the last two fitted values, the residuals are much more spread out.

```r
plotResiduals(mod.q3.1)
```

Figure 8: Residual Analysis



3.3 Part c

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

Solution:

$\hat{y}_i = 2.026 + 0.133 \times I(x_{1i} = \text{“Stage 1”}) + 0.304 \times I(x_{1i} = \text{“Stage 2”}) + 0.774 \times I(x_{1i} = \text{“Stage 3”}) + 0.990 \times I(x_{1i} = \text{“Stage 4”})$

* $\beta_0 = 2.026$: the expected value of log(MFAP4) when the reference category is “Stage 0”

- $\beta_1 = 0.133$: the change in log(MFAP4) when the reference category changes from “Stage 0” to “Stage 1”
- $\beta_2 = 0.304$: the change in log(MFAP4) when the reference category changes from “Stage 0” to “Stage 2”
- $\beta_3 = 0.774$: the change in log(MFAP4) when the reference category changes from “Stage 0” to “Stage 3”
- $\beta_4 = 0.990$: the change in log(MFAP4) when the reference category changes from “Stage 0” to “Stage 4”

```
dat = dat |>
  mutate(Log_MFAP4 = log(MFAP4)) |>
  select(Fibrosis.Stage, Log_MFAP4)
```

```
mod.q3 = lm(Log_MFAP4 ~ Fibrosis.Stage, data = dat)
```

3.4 Part d

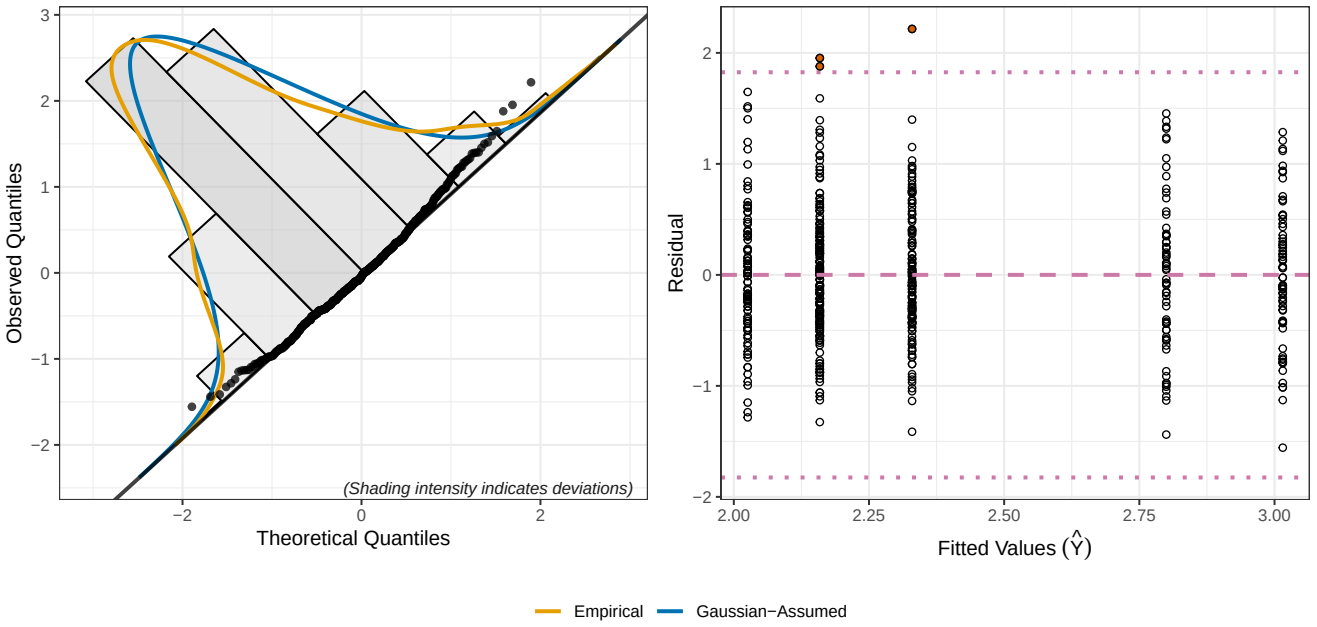
Conduct a residual analysis on the model in part c to demonstrate that the conditions for using regression are met.

Solution:

The log transformation makes the outliers less extreme. The conditions for using regression are met: the residuals follow a Gaussian distribution, and the variance is roughly the same across the five fitted values.

```
plotResiduals(mod.q3)
```

Figure 9: Residual Analysis



3.5 Part e

Fit the model predicting `log(MFAP4)` using `Fibrosis.Stage`, `Age`, and `Gender`. Use the natural log, and not the $\log_2(\cdot)$ recorded in the data file. Write out the resulting estimate model and interpret the coefficients.

Solution:

$$\begin{aligned}\hat{y}_i =& 1.519 + 0.127 \times I(x_{1i} = \text{“Stage 1”}) + 0.213 \times I(x_{1i} = \text{“Stage 2”}) \\ &+ 0.599 \times I(x_{1i} = \text{“Stage 3”}) + 0.829 \times I(x_{1i} = \text{“Stage 4”}) \\ &+ 0.013 x_{2i} \\ &- 0.143 \times I(x_{3i} = \text{“Male”})\end{aligned}$$

Holding other variables fixed,

- $\beta_0 = 1.519$: the expected value of log(MFAP4) when the reference category is “Stage 0”, age is 0, and gender is “Female”. This is not very meaningful because age cannot be 0.
- $\beta_1 = 0.127$: the change in log(MFAP4) when the reference category changes from “Stage 0” to “Stage 1”
- $\beta_2 = 0.213$: the change in log(MFAP4) when the reference category changes from “Stage 0” to “Stage 2”
- $\beta_3 = 0.599$: the change in log(MFAP4) when the reference category changes from “Stage 0” to “Stage 3”
- $\beta_4 = 0.829$: the change in log(MFAP4) when the reference category changes from “Stage 0” to “Stage 4”
- $\beta_5 = 0.013$: the change in log(MFAP4) for every unit increase in age
- $\beta_6 = -0.143$: the change in log(MFAP4) when the reference category changes from “Female” to “Male”

The coefficients decrease after adjustment for simultaneous changes in other predictors.

```
dat2 = dat.mfap4 |>
  mutate(Log_MFAP4 = log(MFAP4)) |>
  select(Fibrosis.Stage, Log_MFAP4, Age, Gender)
```

```
lm(Log_MFAP4 ~ Fibrosis.Stage + Age + Gender, data = dat2)
```

```

Call:
lm(formula = Log_MFAP4 ~ Fibrosis.Stage + Age + Gender, data = dat2)

Coefficients:
          (Intercept)  Fibrosis.StageStage 1  Fibrosis.StageStage 2
              1.51867                0.12686                0.21254
Fibrosis.StageStage 3  Fibrosis.StageStage 4                    Age
              0.59887                0.82855                0.01311
           Gendermale
              -0.14340
```

4 Question 4

4.1 Part a

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$\begin{aligned}x_i &\sim \text{Uniform}(a = 0, b = 10) \\ \epsilon_i &\sim \text{Gaussian}(\mu = 0, \sigma = 1) \\ y_i &= 2 + 1.5 x_i + \epsilon_i \\ n &= 30\end{aligned}$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

Solution:

When the distribution of the residual terms are normal and has a mean of 0, the estimates of the intercept (β_0) and the slope (β_1) follow a Gaussian distribution, centering around the specified intercept and slope.

```
set.seed(8279)

estimate_a = tibble(
  b0 = rep(NA, 1000),
  b1 = rep(NA, 1000)
)

```

```r
for(i in 1:1000) {
  x = runif(30, 0, 10)
  epsilon = rnorm(30, 0, 1)
  y = 2 + 1.5*x + epsilon

  mod = lm(y ~ x)

  estimate_a$b0[i] = coef(mod)["(Intercept)"]
  estimate_a$b1[i] = coef(mod)["x"]

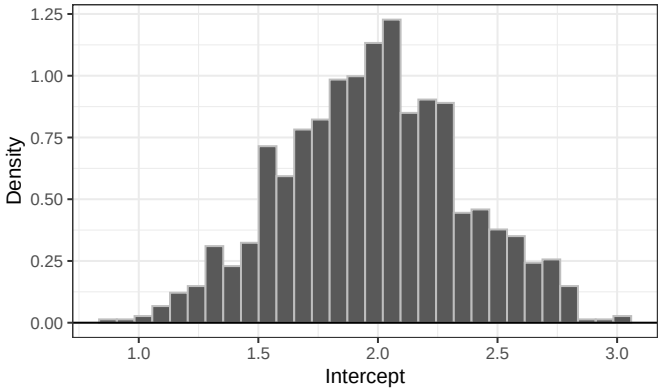
}
```

```r
ggplot(estimate_a) +
  geom_histogram(aes(x = b0, y = after_stat(density)),
                 color = "gray") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Intercept",
       y = "Density")

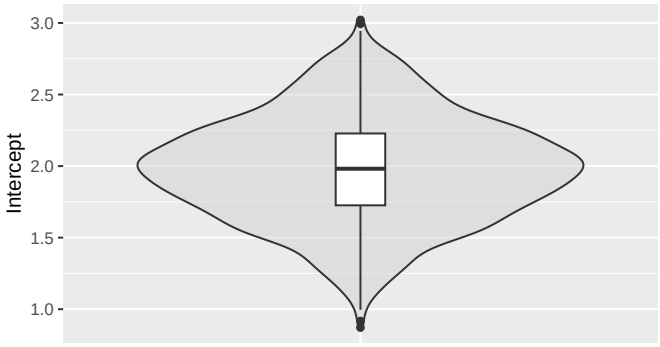
ggplot(estimate_a) +
  geom_violin(aes(x = "", y = b0),
              fill = "lightgrey", alpha = 0.5) +
  geom_boxplot(aes(x = "", y = b0),
               width = 0.1) +
  labs(x = "",
       y = "Intercept") +
  theme(axis.ticks.x = element_blank())
```

Figure 10: Distribution of the intercept estimates

(a) Histogram



(b) Violin plot with boxplot




```r
ggplot(estimate_a) +
  geom_histogram(aes(x = b1, y = after_stat(density)),
                 color = "gray") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Slope",
       y = "Density")

ggplot(estimate_a) +
  geom_violin(aes(x = "", y = b1),
              fill = "lightgrey", alpha = 0.5) +
  geom_boxplot(aes(x = "", y = b1),
               width = 0.1) +
  labs(x = "",
       y = "Slope") +
  theme(axis.ticks.x = element_blank())
```

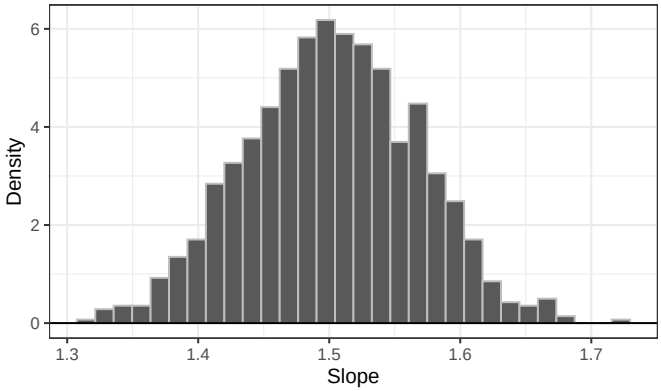
4.2 Part b

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

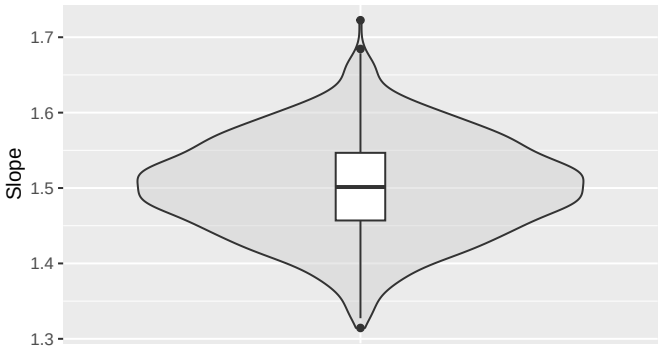
$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Shifted Exponential}(\lambda = 1, \mu = 0) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

Figure 11: Distribution of the slope estimates

(a) Histogram



(b) Violin plot with boxplot



In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

Hint: Use the `tolerance::r2exp()` function with `rate=1` and `shift=-1`.

Solution:

When the distribution of the residual terms is not normal but still centers around 0, the β_0 estimate follows a right-skewed normal distribution, and β_1 estimate follows a Gaussian distribution.

```
set.seed(8279)

estimate_b = tibble(
b0 = rep(NA, 1000),
b1 = rep(NA, 1000)
)

for(i in 1:1000) {
  x = runif(30, 0, 10)
  epsilon = tolerance::r2exp(30, rate = 1, shift = -1)
  y = 2 + 1.5*x + epsilon

  mod = lm(y ~ x)

  estimate_b$b0[i] = coef(mod)["(Intercept)"]
  estimate_b$b1[i] = coef(mod)["x"]
}
```

```
ggplot(estimate_b) +
  geom_histogram(aes(x = b0, y = after_stat(density)),
                 color = "gray") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Intercept",
       y = "Density")

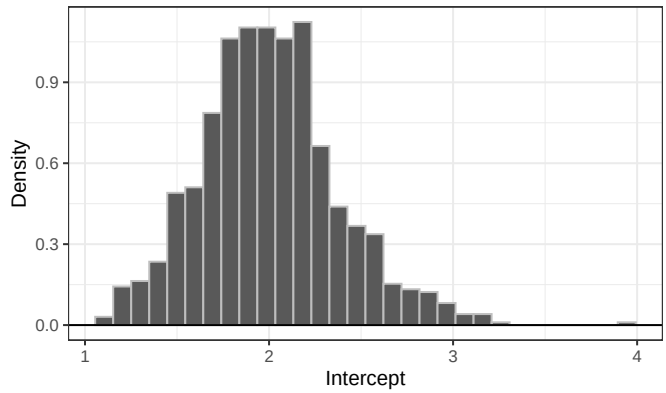
ggplot(estimate_b) +
  geom_violin(aes(x = "", y = b0),
              fill = "lightgrey", alpha = 0.5) +
  geom_boxplot(aes(x = "", y = b0),
               width = 0.1) +
  labs(x = "",
       y = "Intercept") +
  theme(axis.ticks.x = element_blank())
```

```
ggplot(estimate_b) +
  geom_histogram(aes(x = b1, y = after_stat(density)),
                 color = "gray") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Slope",
       y = "Density")

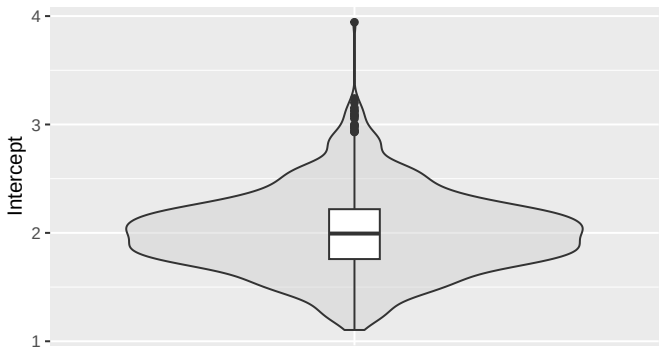
ggplot(estimate_b) +
  geom_violin(aes(x = "", y = b1),
              fill = "lightgrey", alpha = 0.5) +
  geom_boxplot(aes(x = "", y = b1),
               width = 0.1) +
```

Figure 12: Distribution of the intercept estimates

(a) Histogram



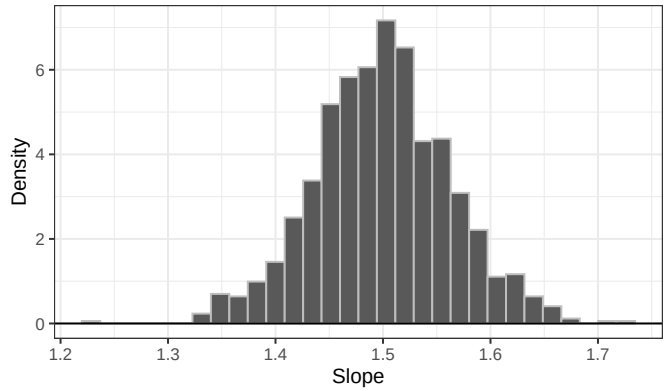
(b) Violin plot with boxplot



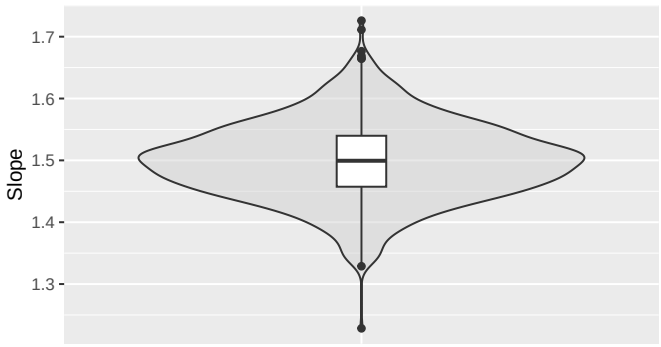

```
  labs(x = "",
       y = "Slope") +
  theme(axis.ticks.x = element_blank())
```

Figure 13: Distribution of the slope estimates

(a) Histogram



(b) Violin plot with boxplot



4.3 Part c

Simulate data to match regression assumptions in a loop that runs $R = 1000$ times.

$$
\begin{aligned}
x_i &\sim \text{Uniform}(a = 0, b = 10) \\
\epsilon_i &\sim \text{Cauchy}(m = 0, s = 1) \\
y_i &= 2 + 1.5x_i + \epsilon_i \\
n &= 30
\end{aligned}
$$

In each iteration, fit a regression line to the data and store the coefficient estimates. Plot the resulting estimates. What do you notice?

Solution:

When the residual terms are drawn from a Cauchy distribution, which has no finite mean or variance and has long tails, they're more likely to be extreme values.

Looking at the medians, β_0 estimate still clusters around 2 and β_1 estimate clusters around 1.5.Looking at Figure 14 and Figure 15, we see some astronomical outliers (below -20000 or above 8000). The lack of constant variance of the Cauchy distribution causes the estimate to be extremely low in precision.

```r
set.seed(8279)

estimate_c = tibble(
b0 = rep(NA, 1000),
b1 = rep(NA, 1000)
)

for(i in 1:1000) {
  x = runif(30, 0, 10)
  epsilon = rcauchy(30, 0, 1)
  y = 2 + 1.5*x + epsilon

  mod = lm(y ~ x)

  estimate_c$b0[i] = coef(mod)["(Intercept)"]
  estimate_c$b1[i] = coef(mod)["x"]
}
```

```r
(median(estimate_c$b0))
```

```
[1] 1.840345
```

```r
(median(estimate_c$b1))
```

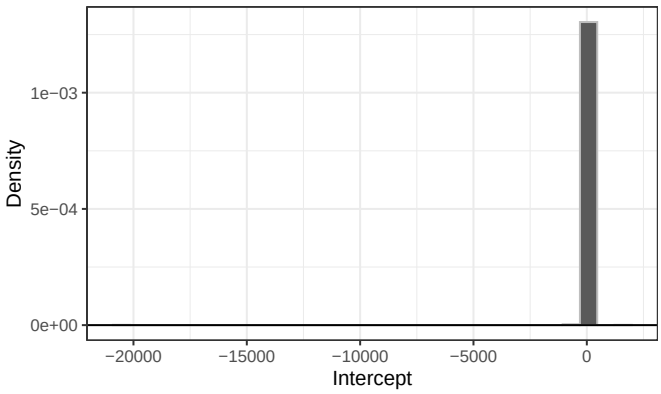
```
[1] 1.526023
```

```r
ggplot(estimate_c) +
  geom_histogram(aes(x = b0, y = after_stat(density)),
                 color = "gray") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Intercept",
       y = "Density")

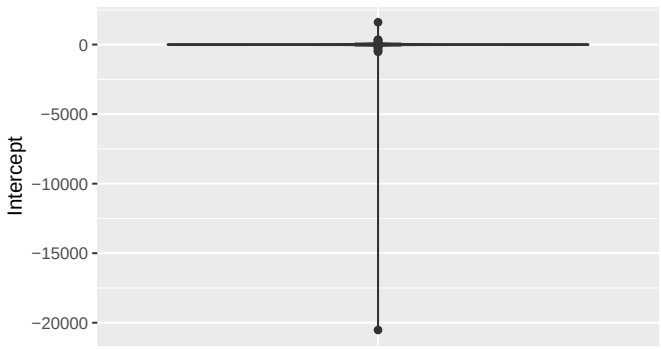
ggplot(estimate_c) +
  geom_violin(aes(x = "", y = b0),
              fill = "lightgrey", alpha = 0.5) +
  geom_boxplot(aes(x = "", y = b0),
               width = 0.1) +
  labs(x = "",
       y = "Intercept") +
  theme(axis.ticks.x = element_blank())
```

Figure 14: Distribution of the intercept estimates

(a) Histogram



(b) Violin plot with boxplot

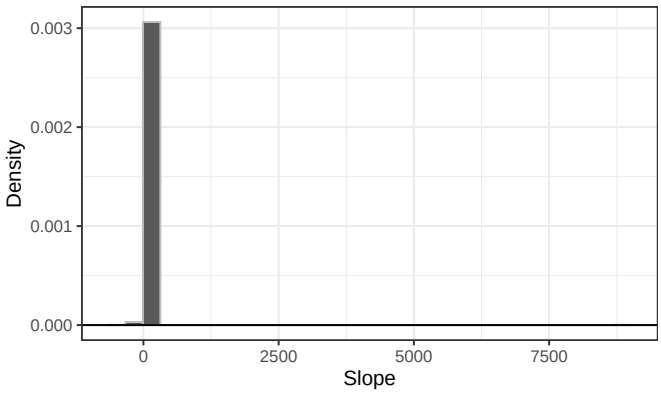



```r
ggplot(estimate_c) +
  geom_histogram(aes(x = b1, y = after_stat(density)),
                 color = "gray") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Slope",
       y = "Density")

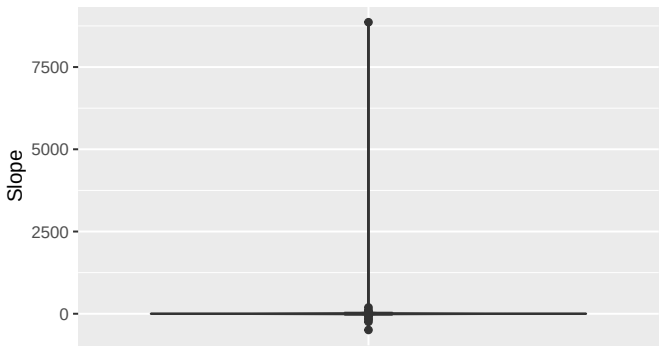
ggplot(estimate_c) +
  geom_violin(aes(x = "", y = b1),
              fill = "lightgrey", alpha = 0.5) +
  geom_boxplot(aes(x = "", y = b1),
               width = 0.1) +
  labs(x = "",
       y = "Slope") +
  theme(axis.ticks.x = element_blank())
```

Figure 15: Distribution of the slope estimates

(a) Histogram



(b) Violin plot with boxplot



References

Bracht, Thilo, Christian Mölleken, Maike Ahrens, et al. 2016. "Evaluation of the Biomarker Candidate MFAP4 for Non-Invasive Assessment of Hepatic Fibrosis in Hepatitis C Patients." *Journal of Translational Medicine* 14 (1): 201.

Cui, Mengyang, and Jian Mao. 2025. "Effect of Low Temperature on Mechanical Properties and Fracture Behavior of 3D Printed Short Carbon Fiber Reinforced Polyether Ether Ketone Composites (CF/PEEK)." *Scientific Reports* 15 (1): 36244.

Long, Jacob A. 2024. *Interactions: Comprehensive, User-Friendly Toolkit for Probing Interactions*. https://doi.org/10.32614/CRAN.package.interactions.